Regulatory Wedge: ChatGPT Access Bans as a Natural Experiment for Open-Source Contribution

Feedback & Improvement Report

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87

/ 100

GATE: FAILED

Reason: polish = 64 (below minimum 70)

1. Score Breakdown

Aggregate score: 86.8/100 | Gate: FAILED | Threshold: 85/100

Identification (30%) **84 / 100**

Credibility of causal design: control group, pre-trends, exogeneity.

Code (20%) **89 / 100**

Scripts run cleanly, outputs exist, SE clustering, robustness checks.

Paper (25%) **97 / 100**

All sections present, references resolve, tables/figures, word count.

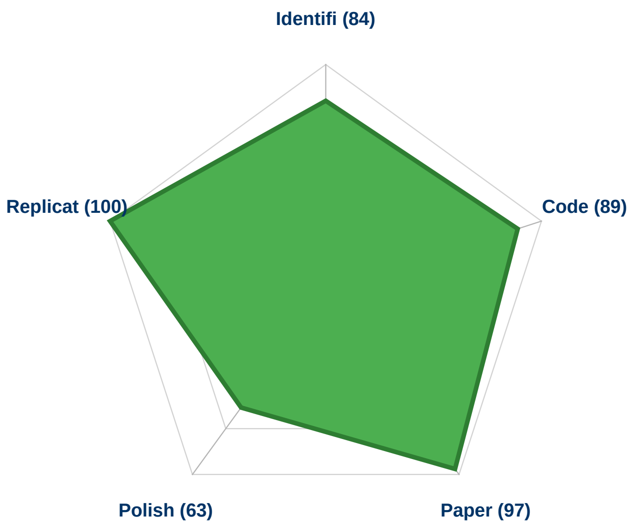
Polish (15%) **64 / 100**

Average referee score from peer review. Hardest to improve.

Replication (10%) **100 / 100**

All scripts reproduce identical outputs when re-run.

Score Profile (radar chart)



What each score means

Identification (84/100):
How credible is the causal claim? A high score means the study has a valid control group, clean pre-trends, and addresses confounders. Studies without a control group (e.g., before-after designs) score below 85. RCTs and strong natural experiments score 90+.

Code (89/100):
Do the scripts run correctly and implement the required analyses? Checks include: all 4 scripts exist and execute, output files are non-empty, standard errors are appropriate, and the referee checklist requirements are addressed in the code.

Paper (97/100):

Is the paper complete and well-structured? Checks include: all sections present (intro through conclusion), references resolve, tables and figures included, word count between 5,000-12,000, and key numbers in the abstract match the actual results.

Polish (64/100):

How did the simulated referees rate the paper? This is the average score from the peer review stage. It reflects how well the paper presents its findings, handles robustness checks, and addresses identification concerns. This is the hardest score to improve because it depends on data quality and research design.

Replication (100/100):

Can the results be reproduced? The pipeline re-runs all scripts and checks that outputs are identical. A score of 100 means perfect reproducibility.

2. Results Diagnostics

Main findings

DiD_persistent_bans: coef = -0.3888 (p=0.1346)

pre_trend_F_test: coef = +3.2650 (p=0.0003) ***

synth_control_italy: coef = +0.0130

Robustness summary

4 of 8 robustness specifications significant (p<0.05).

Key robustness checks

Each row tests if the main result holds under different conditions. *** = p<0.01, ** = p<0.05, * = p<0.10.

Specification	Coef	p-value	Sig
placebo_outcome_n_languages	-0.1664	0.0622	*
placebo_timing_Q2_2021	-0.0023	0.9927	
did_CN_only	-1.3858	0.0000	***
did_RU_only	-0.5486	0.0000	***
exclude_covid_2020	-0.3984	0.0239	**
oecd_only_controls	-0.0255	0.9217	
exclude_small_countries	-0.3808	0.1421	
permutation_test	-0.3888	0.0060	***

3. Data Limitations

Data summary

Rows: 161,922 | Columns: 6 | Structure: repeated-cross-sections

Parallel trends assumption

DiD identification relies on treated and control units following the same trajectory absent treatment. Violated pre-trends bias the estimate.

Confounders concurrent with treatment

Other events at the same time as treatment (wars, policy changes, economic shocks) may confound the estimated effect.

Unresolved referee concerns

The following issues were flagged by referees and may limit publishability: Pre-trend violation (F-test p=0.0003): the positive pre-treatment arc through 2021 that reverses in 2022 coincides with Russia-Ukraine sanctions (Febr; Russia as treated unit: the Russia-Ukraine war (February 2022) and associated IT-sector emigration break Russia's pre-treatment trend nine months befo; Q4 2022 first-treatment timing: ChatGPT launched November 30 ? the final day of Q4 2022. A significant effect in that quarter cannot reflect adoption

4. Validator Results

Code: 12/12 hard, 6/10 soft

X seed_set:00_clean.py: No random seed set

X wild_cluster_bootstrap: No wild cluster bootstrap detected ? recommended for <50 clusters

X referee_checklist_coverage: Referee checklist: 21/53 MUST requirements detected in code (40%)

X referee_checklist_missing: Possibly unimplemented: [DOMAIN] estimation: Italy synthetic control: implement Abadie (2010) SCM via cv; [DOMAIN] estimation: Synthetic DiD (Arkhangelsky et al. 2021) as th

Paper: 4/4 hard, 4/5 soft

X word_count: 2693 words (outside 5000-15000 range)

Integration: 7/7 hard, 11/11 soft

All checks passed.

5. Recommendations

Referee must-address issues

These issues were flagged by simulated referees during peer review.

ISSUE #1 Pre-trend violation (F-test p=0.0003): the positive pre-treatment arc through 2021 that reverses in 2022 coincides with Russia-Ukraine sanctions (February 2022), not ChatGPT. Rambachan-Roth sensitivity analysis is required; verbal dismissal is insufficient.

ISSUE #2 Russia as treated unit: the Russia-Ukraine war (February 2022) and associated IT-sector emigration break Russia's pre-treatment trend nine months before ChatGPT, rendering it an invalid treated unit. Russia must be dropped or its inclusion formally justified with a clean pre-trend.

ISSUE #3 Q4 2022 first-treatment timing: ChatGPT launched November 30 ? the final day of Q4 2022. A significant effect in that quarter cannot reflect adoption and more plausibly captures prior Russia-Ukraine dynamics. Re-estimate with Q1 2023 as the first treatment quarter.

ISSUE #4 OECD-null result (coeff=-0.026, p=0.922): the effect disappears entirely when restricted to economically comparable peers. This is the most informative finding and strongly suggests the baseline result reflects differential development trajectories across heterogeneous controls, not ChatGPT-specific effects. A propensity-score-matched or entropy-balanced control group is required; the OECD null must be treated as the primary robustness test, not a footnote.

ISSUE #5 Single-cluster standard errors for China-only and Russia-only regressions: with one treated cluster, Liang-Zeger cluster-robust SEs break down. The identical SE=0.036 across both regressions is statistically implausible and may indicate a coding error. Wild cluster bootstrap with imposed null is required for all single-treated-cluster estimates.

Score-based recommendations

[HIGH IMPACT] **Improve identification strategy**

Add exogenous variation: natural experiments, policy discontinuities, staggered rollout, or IVs. RCTs score highest.

[MEDIUM] **Improve code quality**

Address referee checklist gaps. Use validated packages (pyfixest, linearmodels). Add missing robustness checks.

[LOW] **Address referee feedback**

Polish improves when identification and code improve. Focus on must-address issues listed above.

6. Score Impact Estimates

Estimated ceiling: ~88 / 100

Current aggregate: 86.8. With the current identification strategy, the practical ceiling is ~88. Reaching 90+ requires RCT or strong natural experiment data.

Improvement estimates

Improvement	Estimated Impact
Stronger identification (RCT or natural experiment)	+10 to +20 identification
Address all referee must-address issues	+5 to +10 polish
Address all SHOULD-address issues	+3 to +5 polish